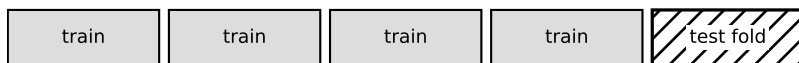


Participants with a usual walk

PhysioNet, three sub-studies: 93 PD, 72 control

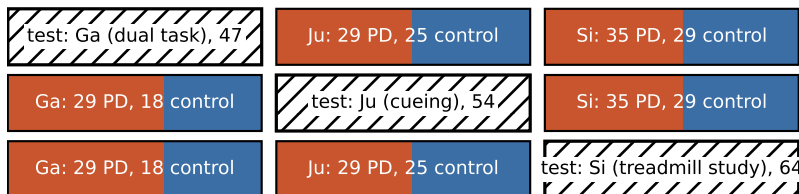
6 timing features per participant
(+ age and sex as covariates)
logistic regression, SVM, random forest;
tuning inside the training fold only

A. Within protocol: 5-fold cross-validation grouped by participant, repeated 20 times



Each fold holds about a fifth of the participants, stratified by group; a participant's recordings are never split between train and test. Reshuffled 20 times; the mean AUC over repeats is reported with its 2.5th to 97.5th percentile.

B. Across protocols: leave one sub-study out



Train on two sub-studies, test on the third. The protocols, recording lengths and ages differ between sub-studies, so this is the primary comparison; the interval is a bootstrap over the held-out participants.

C. Across cohorts: external test



A different research group and hardware; the same stride and swing timings from its pressure insoles or walkway. Run once, after the models are frozen.

PD control held out for testing